



MUSCLES AND SCAFFOLDS

Muscular Systems as Scaffolds

The use of muscular systems as scaffolds in regenerative medicine is an area of active research and development. Muscles have the potential to be used as scaffolds for the regeneration of tissues due to their inherent mechanical properties and ability to support cell growth and tissue formation.

One example of using muscular systems as scaffolds is in the treatment of damaged or diseased heart tissue. Researchers have developed methods for using muscle cells to create a functional, three-dimensional scaffold that can support the growth of new heart tissue. In this approach, muscle cells are harvested from the patient and then seeded onto a scaffold, such as a hydrogel or artificial matrix. The scaffold provides a framework for the cells to grow and differentiate into new heart tissue, which can help to repair the damaged or diseased tissue.

Another example is in the treatment of skeletal muscle injuries, such as those caused by trauma or disease. In this case, muscle cells can be harvested and seeded onto a scaffold, which can then be implanted into the damaged muscle to promote the growth of new, functional tissue.

While the use of muscular systems as scaffolds is still in the experimental stage, it holds great promise for the treatment of a variety of conditions and represents an area of active research and development in the field of regenerative medicine.

3.3.1 Architecture

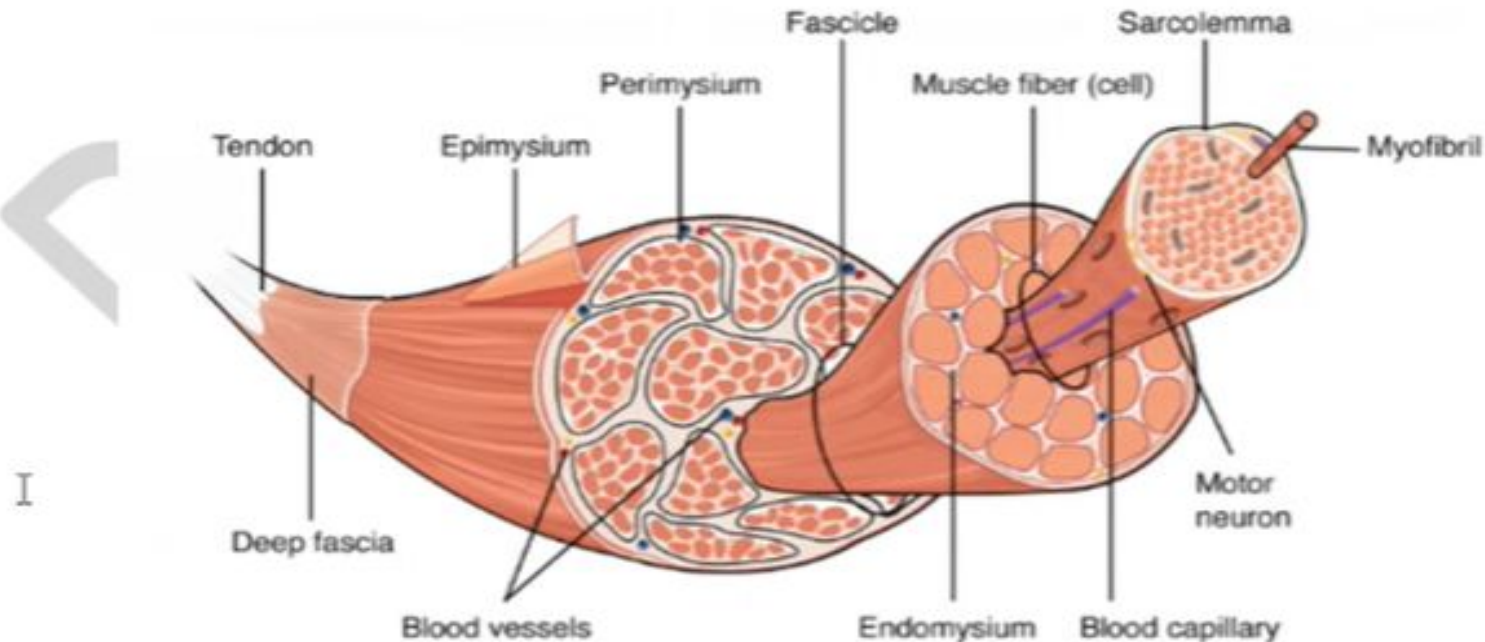


Figure: The Three Connective Tissue Layers: Bundles of muscle fibers, called fascicles, are covered by the perimysium. Muscle fibers are covered by the endomysium.

Inside each skeletal muscle, muscle fibers are organized into bundles, called fascicles, surrounded by a middle layer of connective tissue called the perimysium. This fascicular organization is common in muscles of the limbs; it allows the nervous system to trigger a specific movement of a muscle by activating a subset of muscle fibers within a fascicle of the muscle. Inside each fascicle, each muscle fiber is encased in a thin connective tissue layer of collagen and reticular fibers called the endomysium. The endomysium surrounds the extracellular matrix of the cells and plays a role in transferring force produced by the muscle fibers to the tendons.

Inside the muscle fibers, there are tiny structures called myofibrils. Myofibrils are made up of smaller units called sarcomeres, which are responsible for muscle contraction.

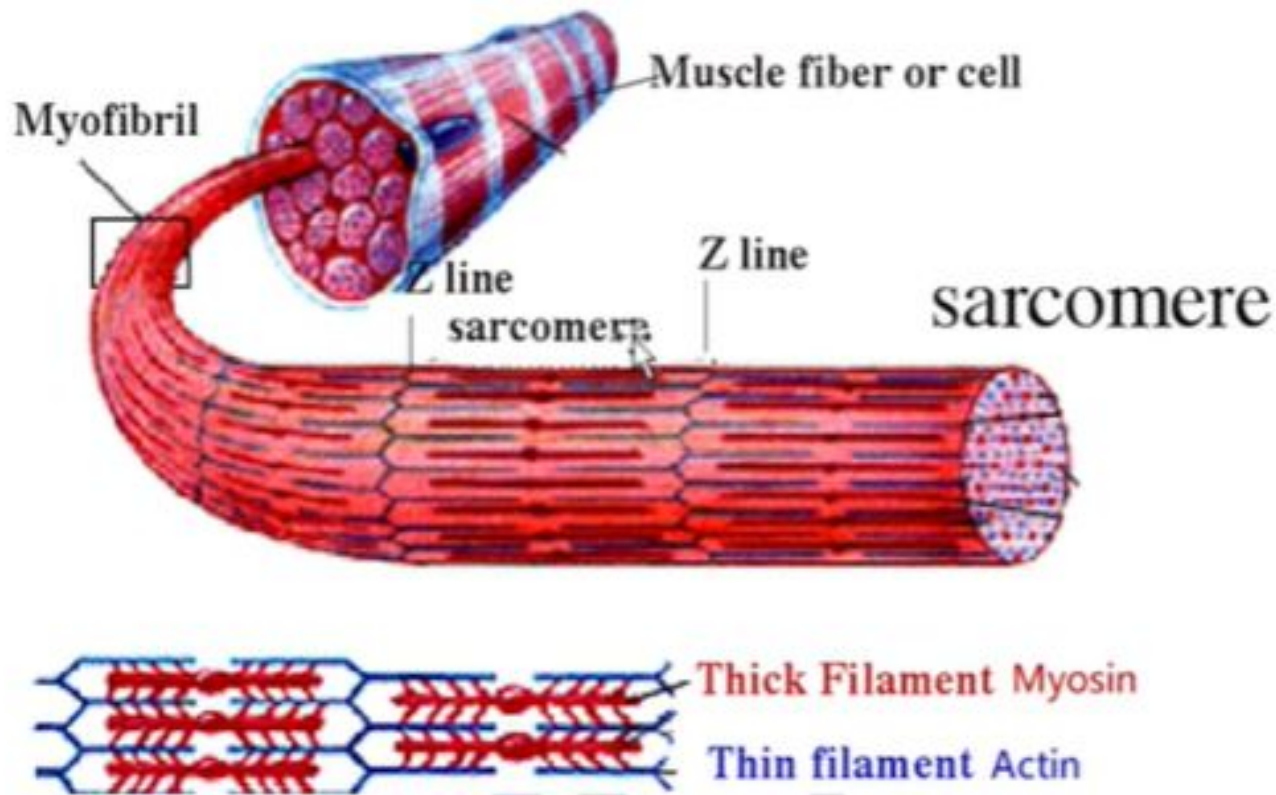


Figure: Representing the sarcomere

Sarcomeres contain thin (Actin) and thick filaments (Myosin) that work together to make the muscle fibers contract. Each muscle fiber is surrounded by a protective layer called endomysium. Multiple muscle fibers are grouped together into bundles called fascicles. Fascicles are surrounded by another layer of connective tissue called perimysium.

All the fascicles together make up the entire muscle, which is surrounded by a layer called epimysium. The muscle also has a special membrane called the sarcolemma, which protects the muscle fiber. Inside the muscle fiber, there are small tunnels called T-tubules that help transmit signals for muscle contraction. Muscles work through the coordination of motor units, which consist of a motor neuron and the muscle fibers it controls. This architecture allows muscles to generate force, move our bodies, and perform various activities.

Mechanisms

The mechanism of how the muscular system can be used as a scaffold in regenerative medicine involves the use of muscle cells and a scaffold to support the growth and regeneration of new tissue.

The method of growing muscle tissue using hydrogel or artificial scaffold is explained below:

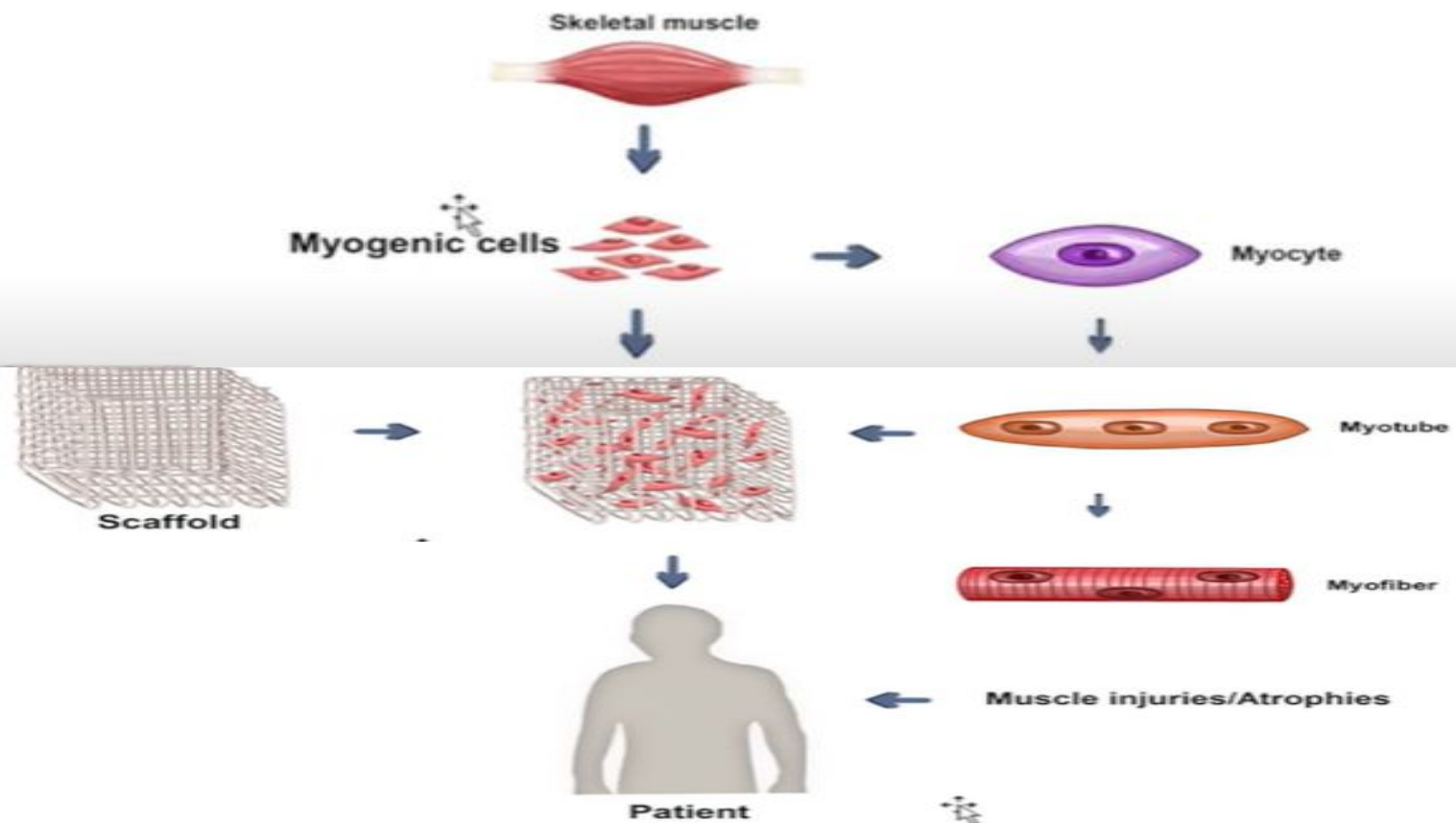


Figure: Representing the muscle tissue growth using hydrogel or artificial scaffold

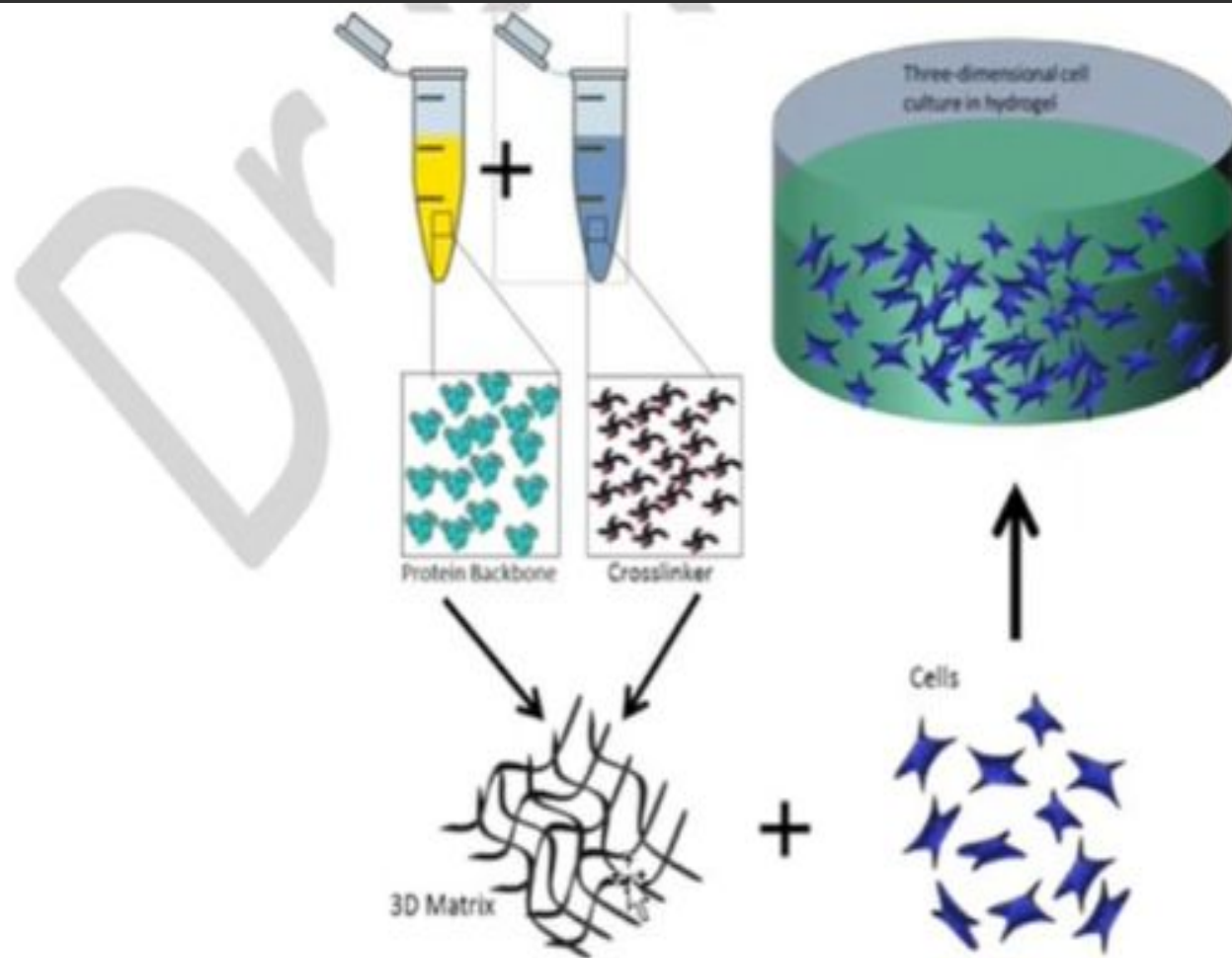


Figure: Representing the formation of polymer based scaffold and cell culture

- Harvesting of muscle cells: Muscle cells are typically obtained from the patient and then isolated and expanded in culture.
- Seeding onto scaffold: The muscle cells are then seeded onto a scaffold, such as a hydrogel or artificial matrix. The scaffold provides a framework for the cells to grow and differentiate into new tissue.
- Cell differentiation and tissue formation: Once the cells are seeded onto the scaffold, they undergo differentiation, in which they change into specific cell types, such as muscle cells or heart cells. The cells also begin to organize and form new tissue, such as heart tissue or skeletal muscle tissue.
- Implantation into patient: The scaffold and cells are then implanted into the patient to promote the growth of new, functional tissue.

3.3.3 Muscle Cells as Scaffold

Muscle cells can be used as a scaffold for tissue generation by removing the living cells from the muscle tissue, leaving behind the structure known as the extracellular matrix (ECM). This decellularized muscle scaffold provides a framework that can guide and support the growth of new tissues.

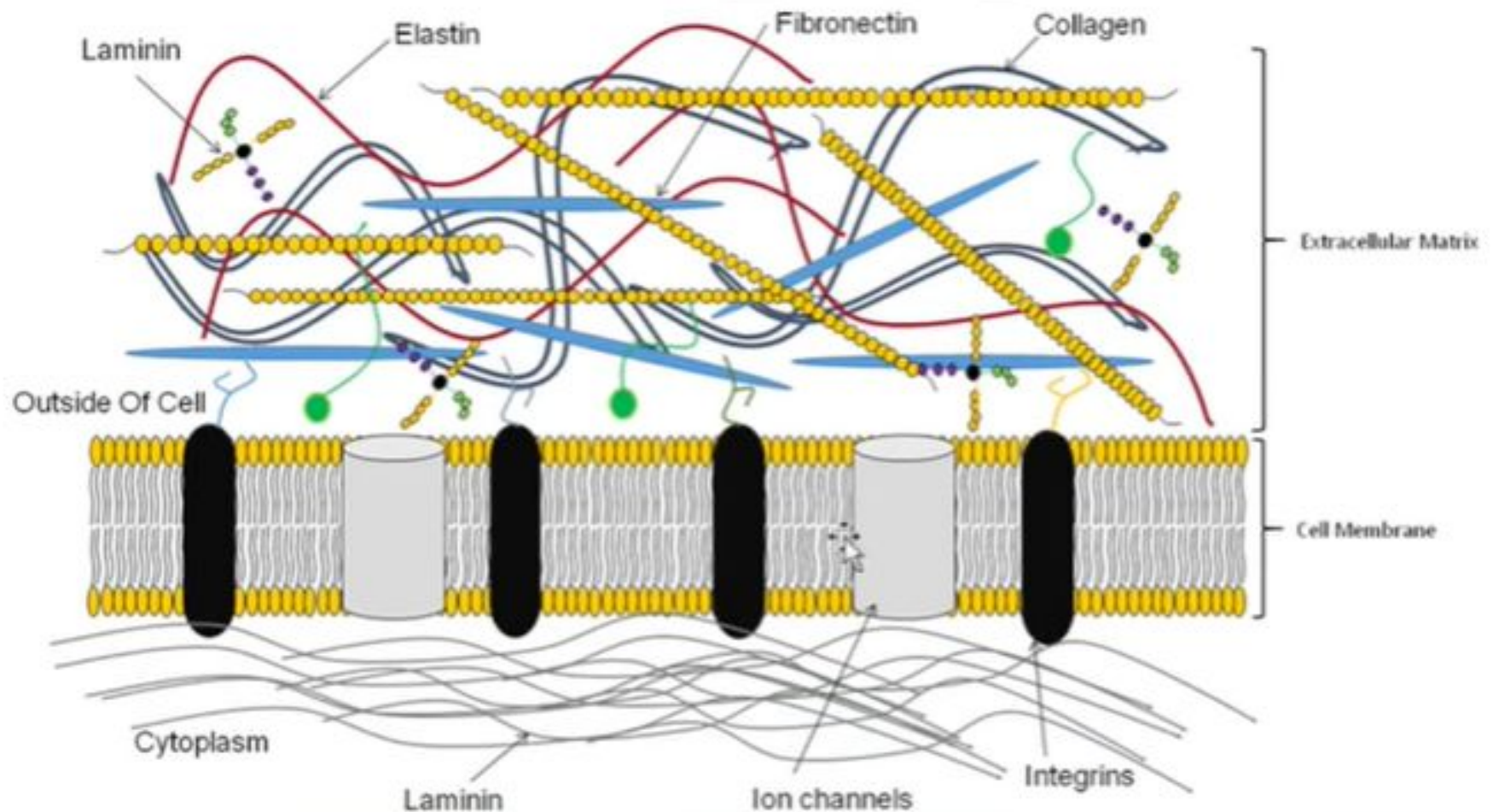


Figure: Representing muscle scaffold for tissue growth

The Process

- **Harvesting muscle tissue:** A small sample of muscle tissue is taken, typically from a donor or an animal model.
- **Cell removal:** The living cells within the muscle tissue are removed using a process called decellularization. This involves treating the tissue with specific chemical solutions or enzymes that break down and wash away the cellular components, while **preserving** the ECM.
- **ECM scaffold:** The remaining ECM, which forms the structure of the muscle scaffold. It consists of proteins, such as collagen and elastin, and other molecules that provide support and signals for tissue growth.
- **Seeding cells:** The decellularized muscle scaffold is then seeded with desired cells. These can be stem cells or specialized cells relevant to the type of tissue being regenerated. The cells are introduced onto the scaffold, allowing them to attach and populate the structure.
- **Tissue growth:** Over time, the seeded cells proliferate and differentiate, meaning they multiply and transform into specific cell types required for the desired tissue. The ECM scaffold guides the cells' growth, providing physical support, and biochemical cues to influence their behavior.
- **Tissue integration:** As the cells continue to grow, they populate the scaffold and form new tissue. The new tissue integrates with the surrounding native tissue, gradually replacing the decellularized scaffold with functional, regenerated tissue.

By utilizing the decellularized muscle scaffold, the process of tissue generation takes advantage of the existing three-dimensional architecture and mechanical properties of the muscle. This approach has the potential to address challenges in tissue engineering, such as creating a suitable environment for cell growth, promoting vascularization, and facilitating functional integration of regenerated tissues.

3.3.4 Bioengineering Solutions for Muscular Dystrophy

Normal biceps Muscular dystrophy



Figure: Representing normal muscle and muscular dystrophy

Muscular dystrophy is a group of genetic disorders that result in progressive weakness and degeneration of the skeletal muscles, which are responsible for movement. The disorders are caused by mutations in genes that encode proteins needed for muscle function. The most common type of muscular dystrophy is Duchenne muscular dystrophy, which typically affects young boys and leads to severe disability by early adulthood. Other forms of the disease include Becker muscular dystrophy, limb-girdle muscular dystrophy, and facioscapulohumeral dystrophy, among others.

Duchenne muscular dystrophy (DMD) usually appears early in childhood **between the** ages of 2 and 3. DMD primarily affects boys but can affect girls in rare cases. The primary symptom of DMD is muscle weakness that begins in the muscles close to the body and later affects muscles in the outer limbs.

Becker muscular dystrophy typically becomes apparent between the ages of 5 and 15. It is similar to Duchenne MD, except that it progresses slower and symptoms begin to appear later. Boys are primarily affected by Becker MD. Becker MD causes muscle loss that begins in the hips and pelvic area, thighs, and shoulders.

The age of onset of limb-girdle muscular dystrophy is highly varied, ranging from early childhood to later adulthood. The disease is characterized by muscle weakness and atrophy of the muscles of the hip and shoulder areas (the limb girdles).

Facioscapulohumeral dystrophy (FSHD) typically appears before the age of 20, but can appear later in adulthood or even in childhood in both males and females. FSHD affects the muscles of the face, around the shoulder blades, and in the upper arms.

There is currently no cure for muscular dystrophy, but various treatments can help manage symptoms and slow the progression of the disease. These may include physical therapy, assistive devices, orthopedic surgery, and medication to manage muscle spasms and pain. In some cases, genetic therapy and stem cell transplantation are also being explored as potential treatment options.

It's important for individuals with muscular dystrophy to work closely with a healthcare team that includes specialists in neurology, rehabilitation medicine, and orthopedics, to develop a comprehensive care plan that meets their specific needs.

Bioengineering solutions for muscular dystrophy

Bioengineering solutions for muscular dystrophy aim to improve the lives of individuals affected by the disease by addressing the underlying genetic mutations and muscle weakness. Some of the approaches being explored include:

- **Gene therapy:** This involves delivering a functional copy of the missing or mutated gene to the affected muscle cells. The goal is to restore the production of the missing protein and improve muscle function.
- **Stem cell therapy:** This involves using stem cells to replace the damaged muscle cells and promote repair and regeneration of the muscle tissue. Stem cells can be taken from the patient's own body (autologous stem cells) or from a donor (allogenic stem cells).
- **Exoskeleton technology:** This involves using wearable devices, such as robotic exoskeletons, to support and enhance the movement of individuals with muscular dystrophy. The devices use motors and sensors to mimic the movements of the wearer and help improve mobility.
- **Tissue engineering:** This involves using a combination of materials, such as scaffolds and growth factors, to promote the growth and repair of muscle tissue. The goal is to create functional muscle tissue that can replace the damaged tissue in individuals with muscular dystrophy.

These approaches are still in the early stages of development, but hold promise for the future treatment of muscular dystrophy. Clinical trials and further research are needed to determine the safety and efficacy of these therapies.

SKELLETAL SYSTEM AS SCAFFOLDS

Skeletal Systems as Scaffolds

Skeletal System|

The skeletal system of human beings refers to the framework of bones, joints, and connective tissues that provide structure, support, and protection to the body.

The key components and functions of the skeletal system are:

Bones: The human body consists of 206 bones that vary in size and shape. Bones are composed of hard and dense connective tissue that provides strength and support. They serve as the anchor points for muscles, protect internal organs, and store minerals like calcium and phosphorus.

Cartilage: Cartilage is a flexible connective tissue found in certain joints and structures such as the ears and nose. It acts as a cushion between bones, reducing friction and absorbing shock.

Ligaments: Ligaments are tough bands of fibrous tissue that connect bones to other bones in joints, providing stability and preventing excessive movement.

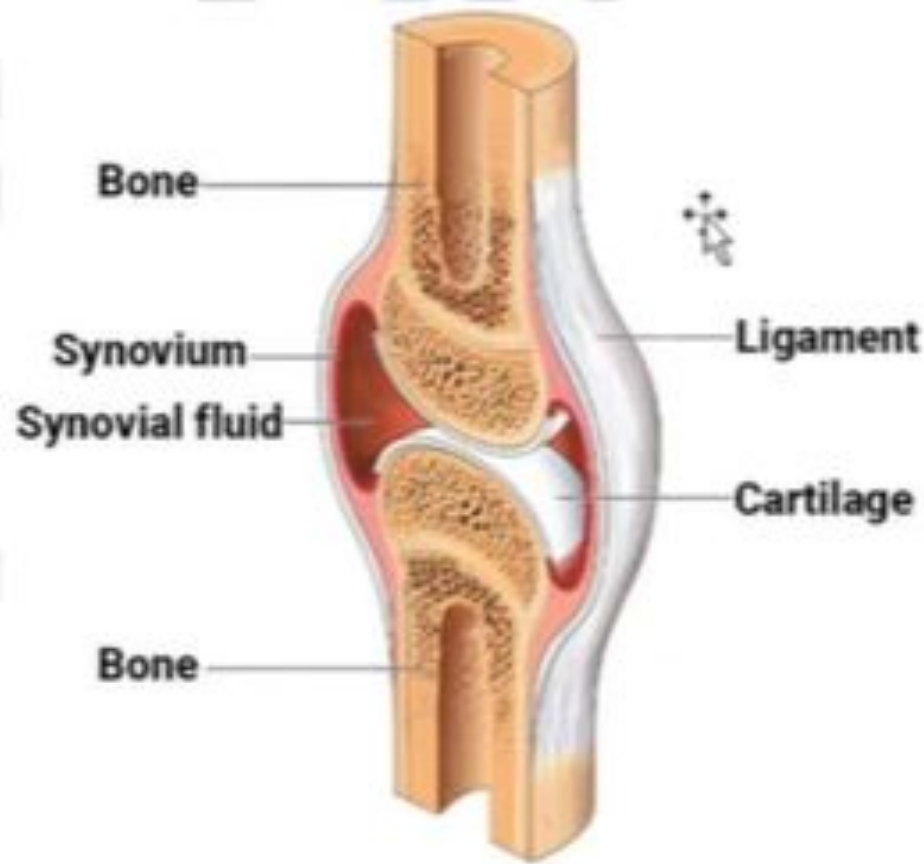


Figure: Representing bone, cartilage, ligament

Tendons: Tendons are strong fibrous tissues that connect muscles to bones, enabling movement by transmitting the force generated by muscles.

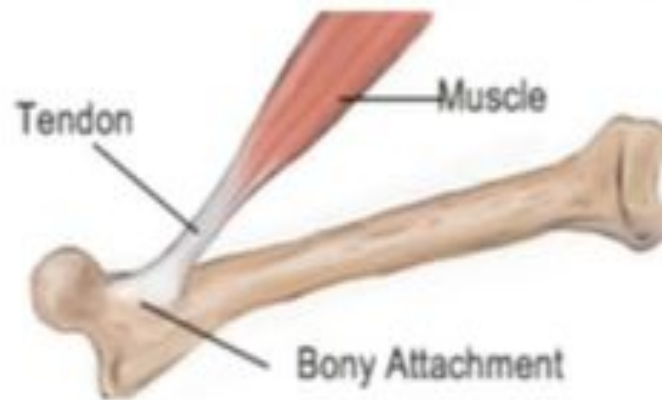


Figure: Representing tendon

Axial Skeleton: The axial skeleton forms the central axis of the body and includes the skull, vertebral column, and ribcage. The skull protects the brain, and the vertebral column (spine) supports the body's weight and houses the spinal cord. The ribcage encloses and protects the heart, lungs, and other thoracic organs.

Appendicular Skeleton: The appendicular skeleton comprises the bones of the limbs and the shoulder and pelvic girdles. The upper limbs (arms) consist of the humerus (upper arm bone), radius and ulna (forearm bones), and the hand bones. The lower limbs (legs) include the femur (thigh bone), tibia and fibula (lower leg bones), and the foot bones. The shoulder and pelvic girdles attach the limbs to the axial skeleton.

Appendicular Skeleton

Axial Skeleton

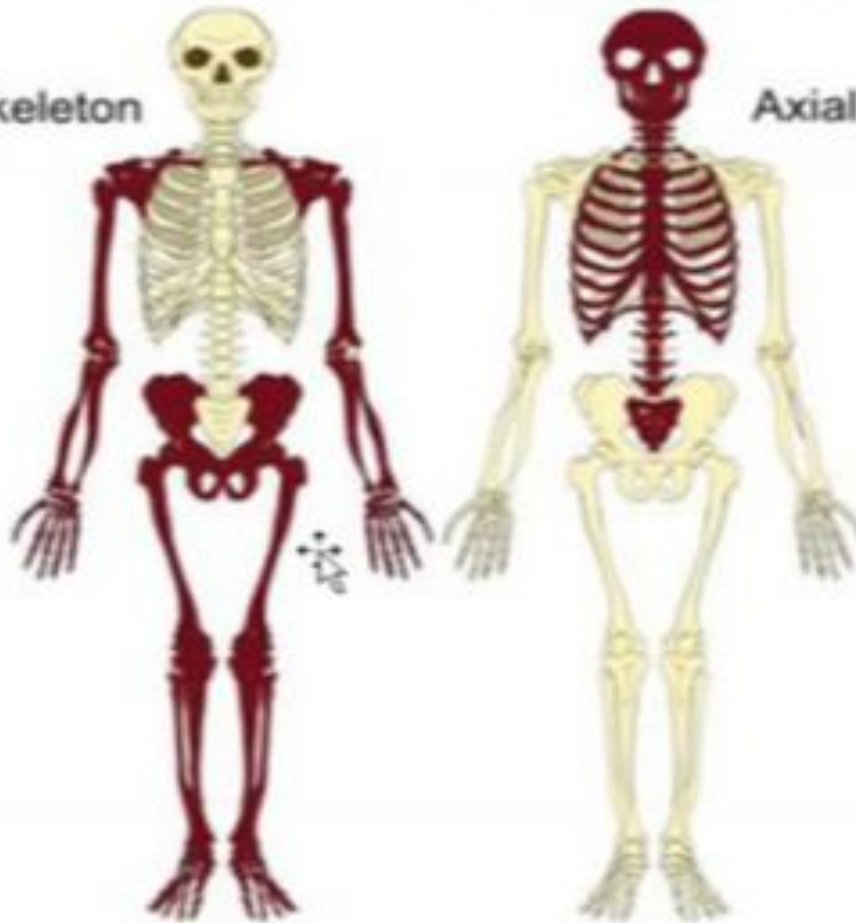


Figure: Representing axial and appendicular skeleton

Joints: Joints are the points where bones meet and allow for movement. There are different types of joints, including hinge joints (e.g., elbow and knee) that enable bending and straightening, ball-and-socket joints (e.g., hip and shoulder) that allow for a wide range of motion, and pivot joints (e.g., between the atlas and axis vertebrae) that allow rotational movement.

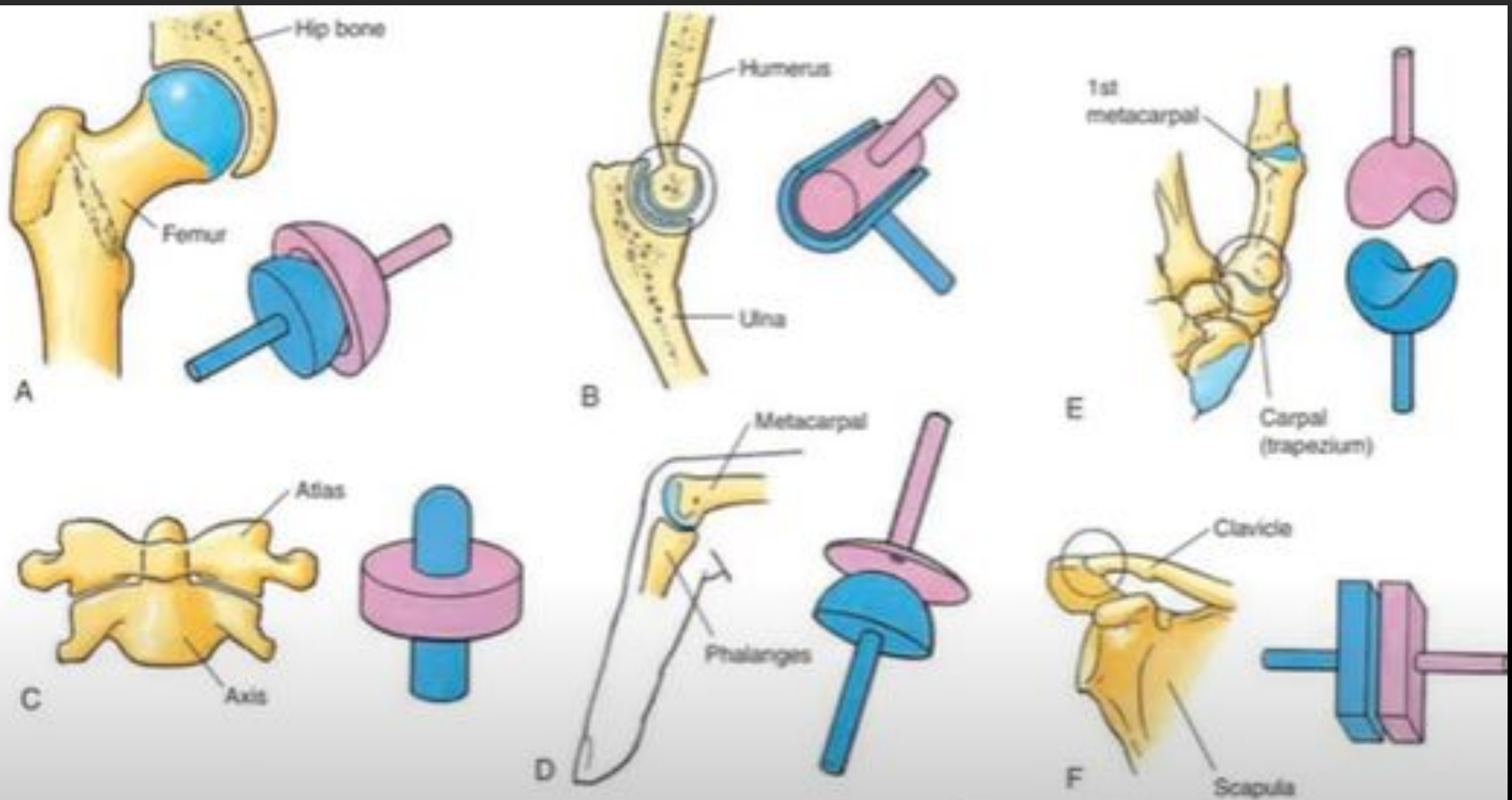


Figure: Representing various skeletal joints A) Ball and socket, B) Hinge, C) Pivot, D) Ellipsoidal, E) Saddle, and F) Glider or planar

The skeletal system works in conjunction with muscles, tendons, and ligaments to allow for movement, protect internal organs, support the body's weight, and provide a structural framework for the body.

3.4.2 Skeletal System as Scaffold

The skeletal system can be used as a scaffold for tissue growth in certain applications. Scaffold-based tissue engineering is a field that aims to create artificial scaffolds to support the growth and regeneration of tissues and organs. In some cases, the natural structure of the skeletal system can serve as a scaffold or template for tissue engineering purposes.

For example, bone tissue engineering often involves the use of scaffolds to facilitate the repair and regeneration of bone defects or injuries. Synthetic or natural biomaterial scaffolds, designed to mimic the properties of bone, can be used to fill the void left by a bone defect. The scaffold provides a three-dimensional structure that supports the attachment, proliferation, and differentiation of cells involved in bone regeneration. Over time, the scaffold can be replaced by newly formed bone tissue, resulting in the restoration of bone structure and function.

In addition to bone tissue engineering, the skeletal system has also been explored as a scaffold for other tissues. For instance, researchers have investigated using decellularized bone or cartilage scaffolds as templates for the regeneration of other tissues like muscle, blood vessels, or nerves. The existing extracellular matrix and structure of the skeletal system can provide a framework for cells to populate and guide tissue growth.

However, it's important to note that using the skeletal system as a scaffold for tissue growth requires careful consideration and modification to match the specific requirements of the target tissue. Additional steps, such as surface modifications, incorporation of bioactive molecules, or cell seeding, may be necessary to optimize the scaffold's effectiveness for promoting tissue regeneration.

Though the skeletal system has potential as a scaffold for tissue growth, successful application requires further research, customization, and integration with tissue engineering strategies specific to the desired tissue type.

3.4.3 Bioengineering Solutions for Osteoporosis

Osteoporosis

Osteoporosis is a condition that weakens the bones and makes them more likely to break (fracture), especially the bones in the hip, spine, and wrist. It occurs when the body loses bone mass and density more quickly than it can be replaced, leading to fragile bones that are prone to fracture.

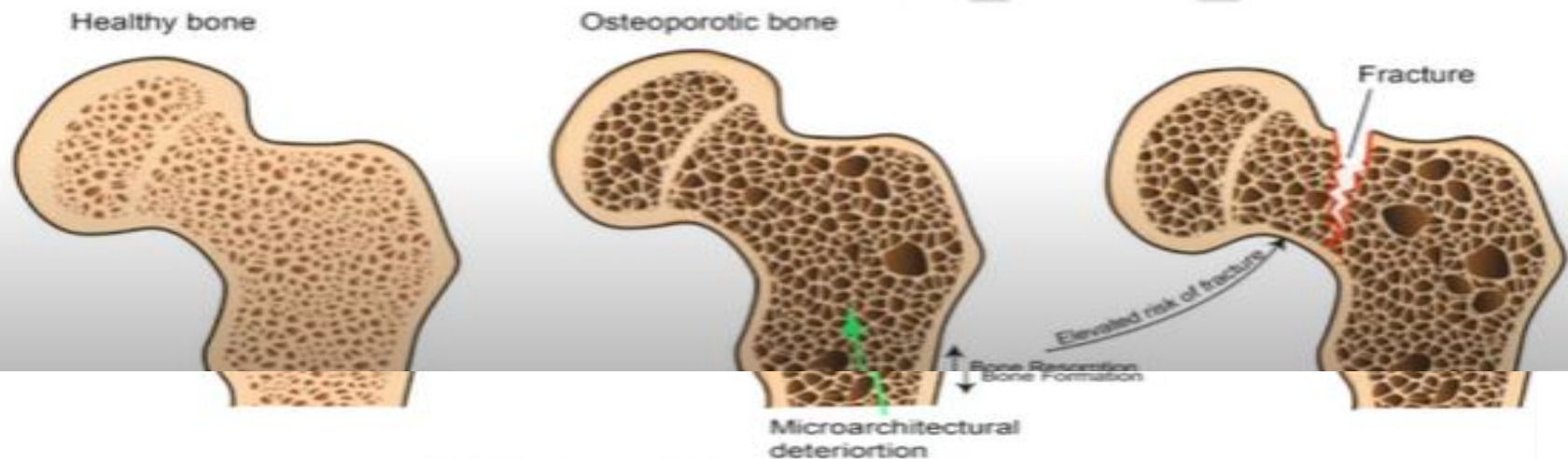


Figure: Representing healthy bone and osteoporotic bone

Osteoporosis is a common condition, especially among older women, and it can increase the risk of falls and fractures, which can result in significant pain and disability. Risk factors for osteoporosis include being female, older age, having a family history of the condition, smoking, drinking excessive amounts of alcohol, being thin or having a small body frame, and having a low calcium intake.

Treatment for osteoporosis aims to slow down bone loss, prevent fractures, and treat fractures if they occur. Some of the treatments include:

- **Medications:** Bisphosphonates, denosumab, and teriparatide are some of the medications that can slow down bone loss and reduce the risk of fractures.
- **Calcium and Vitamin D supplementation:** Calcium and Vitamin D are essential for healthy bones, and taking supplements can help maintain bone mass.
- **Exercise:** Weight-bearing and resistance exercises can help improve bone density and reduce the risk of fractures.

- Lifestyle changes: Quitting smoking, reducing alcohol consumption, and eating a healthy diet that includes enough calcium and Vitamin D can help maintain healthy bones.

It's important to work closely with a healthcare provider to develop a comprehensive treatment plan for osteoporosis, as the right approach may vary depending on the individual's specific needs and medical history.

Bioengineering solutions for osteoporosis

Bioengineering solutions for osteoporosis aim to improve bone health and prevent fractures. Some of the approaches being explored include:

- Tissue engineering: This involves using scaffolds and growth factors to stimulate the growth of new bone tissue and promote the repair of damaged bones. The goal is to create functional bone tissue that can replace the lost bone mass and density in individuals with osteoporosis.
- Stem cell therapy: This involves using stem cells to replace the damaged bone cells and promote the repair and regeneration of bone tissue. Stem cells can be taken from the patient's own body (autologous stem cells) or from a donor (allogenic stem cells).

Biomaterials: This involves using synthetic or natural materials to replace or augment damaged bone tissue. Biomaterials can be designed to mimic the properties of natural bone and promote the growth of new bone tissue.

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- Gene therapy: This involves delivering a functional copy of a gene involved in bone growth and repair to the affected bone cells. The goal is to restore the production of the missing protein and improve bone health.

These approaches are still in the early stages of development, but hold promise for the future treatment of osteoporosis. Clinical trials and further research are needed to determine the safety and efficacy of these therapies.

In addition, traditional treatments for osteoporosis, such as medication, exercise, and lifestyle changes, will likely continue to play an important role in preventing fractures and maintaining healthy bones in individuals with osteoporosis.